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Self sovereign identity in e-governance: blockchain solutions for fintech compliance and citizen-centric financial services

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With the growing technology, population and public services, there is a need for efficient transactions with safety and security in e-governance applications. This study proposes a blockchain supported e-governance digital identity management framework incorporated with gig-economy strategy and service innovation tools that could enhance the existing e-governance process. The proposed conceptual model is validated through expert interviews and fuzzy logic. The process is comprehensive and involves applying the opinions of experts, as well as confirming the rules used for the fuzzy inference system through a questionnaire to obtain a more profound understanding of opinions. The result of the fuzzy logic algorithm indicates that to ensure the individuals' identity is safe and secure, the e-governance applications must integrate blockchain technology, and to make identity management more efficient, there needs to augmentation of service innovation and the gig-economy model. Based on the result of this study, the academicians need to focus on new theoretical and conceptual frameworks in enhancing the usage of blockchain technology integrated with the latest technologies such as AI, ML and Metaverse. The government needs to understand that bringing innovative service models and incorporating the gig economy could help in determining secure and cost-effective transactions, and can be used for framing the models in other industries. The proposed model yielded the highest efficiency scores and better results compared to the existing models in the literature in terms of enhanced productivity and performance. Compared to other studies, this study emphasizes the importance of managing resources through the gig economy, and frequent innovations with secured blockchain would result in enhanced transactions.

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Introduction

Technology transformation is fast occurring globally across sectors, leading to a competitive business environment. Every industry, irrespective of the type of business, has an objective to be an early mover, and hence, they diversify their business strategies. On the other hand, customers expectations and requirements are marching towards personalized comfort without compromises. Every manager aspires to sustain for a longer period. In that case, to sustain itself as a market leader among competing organizations, it has to focus on innovation in technological and non-technological aspects, and hence there evolved co-creation among researchers, academicians, and practitioners evolved. However, this is not the situation with government organizations and agencies, as their motive is to provide public welfare schemes to the masses at a lower cost.

Most governments worldwide have successfully implemented their public welfare schemes using Information and Communication Technologies (ICT) termed E-Governance (Ojo, 2014; Basu et al., 2023a). The success of providing services such as healthcare, education, social justice, food safety and security, first aid and prevention of disease, policies and legal support etc. in developed countries has motivated the governments in developing countries to provide similar services and enhance their quality for their citizens. According to Dawes (2008) and Business Jargons, (2019), E-Governance is a bureaucratic process of providing simple, moral, accountable, responsive, and transparent governance to the citizens, businesses, and others that helps in the reduction of corruption and at the same time provides services to the masses. Further, the politicians and government employees influence can be reduced through online transactions and comfortable with record keeping. For citizens, the technology allows them to access the service at their convenience, provide deeper knowledge, access from any location, and multitask without affecting the quality at less cost, resource, and manpower (Kumar et al., 2021; Raza, 2024; Nair et al., 2023). E-governance application also helps the government in managing the economic activities at macro and micro-level environments, from fiscal to monetary policies, social upliftment, increase in individual per capita income, lifestyle, environment and wildlife management. Specifically, it helps in the reduction of costs and exclusively provides personalized services (Dasgupta, 2021; Basu et al., 2023b).

However, the usage of electronic communication technologies such as the internet and other digital tools for enhancing the effectiveness and efficiency of government operations and public services has several gaps to be addressed, especially concerns like privacy and security when it comes to individuals' digital identity management (Basu, 2004; Wang, 2024). One of the key drawbacks of technology, especially in developing nations, is the digital divide, the availability of technological devices or knowledge to very few people and not others. This creates a situation where only a certain set of the population can access the services online, thereby causing a hindrance to others from participating in democratic processes. Further, the usage of technology in governance could also increase the risk of cyberattacks and data breaches, for which governments must be cautious in the security and protection of sensitive information. One additional requirement is the digital literacy of the citizens. Appropriate knowledge to access and use the government-launched services. Lack of knowledge and compatibility leads to the exclusion from the e-governance benefits, especially those for the unfortified groups like the elderly, low-income and people in poverty (Bwalya, 2009). For example, in developing countries like India, where the population is a major concern, not having an appropriate, adequate infrastructure is also a serious concern. To create and ensure trust among the citizens, the government has to be

accountable and incorporate suitable infrastructure like providing high-speed internet, advanced hardware and software, electricity, and be updated with the latest technology and provide awareness continuously.

There are numerous technologies and software introduced and managed to address the day-to-day requirements. However, academics and researchers favored certain technologies compared to others by conducting deeper research and industrial suggestions (Wang and Chen, 2024). For example, very few literature works discuss blockchain technology (a decentralized transaction database mechanism since its emergence in 2008) compared to artificial intelligence, machine learning, metaverse etc. Blockchain technology was originally invented to support the decentralized financial transaction known as cryptocurrency named Bitcoin; however later it has been adopted for various other applications (Alla et al., 2018; Rana et al., 2022; Nair et al., 2023). Three generations of blockchain technology have been identified by Efanov and Roschin, (2017) as namely: (1) digital currency, (2) digital economy and (3) digital society. The first generation is the digital currency, which is the platform of underlying protocols and digital contracts. The second generation is the digital economy, whereby a variety of economic and financial payments, transfers and transactions occur, such as loans, mortgages, stocks, bonds, derivatives, as well as legal instruments (Nuryyev et al., 2021). It also includes an additional feature of computer-assisted smart contracts, where the related parties to the contracts can automatically make transparent payments as per the contract without any manual form filings. The latest generation is the digital society which refers to the wide use of applications beyond money, currency, commerce, financial instruments, or any other economic activity instead it includes art, health, science, education, governance and other aspects of communication and culture (Swan, 2015; Kaur et al., 2022; Manohar et al., 2023; Wang, 2023). Blockchain has a tremendous success record concerning the applicability of digital identity database management. Digital Identity Management (DIM) is the process of managing and maintaining the individuals' complete information recorded/collected from various sources, further processing, storing and retrieval for future usage securely and safely in digital form (Samir, 2021).

Though blockchain technology is looked at for its benefits, there are also some negative sides. The major drawbacks are implementation at a higher level, interoperability, alignment with other technologies and scalability. Several recent studies (Liu et al., 2023; Yusuf et al., 2023; Sikder, 2023) have also emphasized that there is a need for a secure platform for digital governance to deliver public services, as government agencies and departments would be sharing and using sensitive data. Taking this as a motive, this study examines options in integrating innovations and gig-economy practices in existing blockchain-enabled digital identity management. The paper proposes that adopting/incorporating would enhance the trust among the customers. This study proposes an enhanced Blockchain supported e-governance digital identity management framework (DIMF) that helps ensure the confidentiality of individual information. Further, the study also helps to understand how Gig-Economy Implementation and Service Innovation Models could increase the productivity and efficiency of existing e-governance processes.

Review of literature and conceptual connections

Overview. The literature review for this study focuses on keywords such as blockchain technology, e-governance, digital identity management, privacy concern and technology transformation based on content analysis proposed by Ding, (2019). The

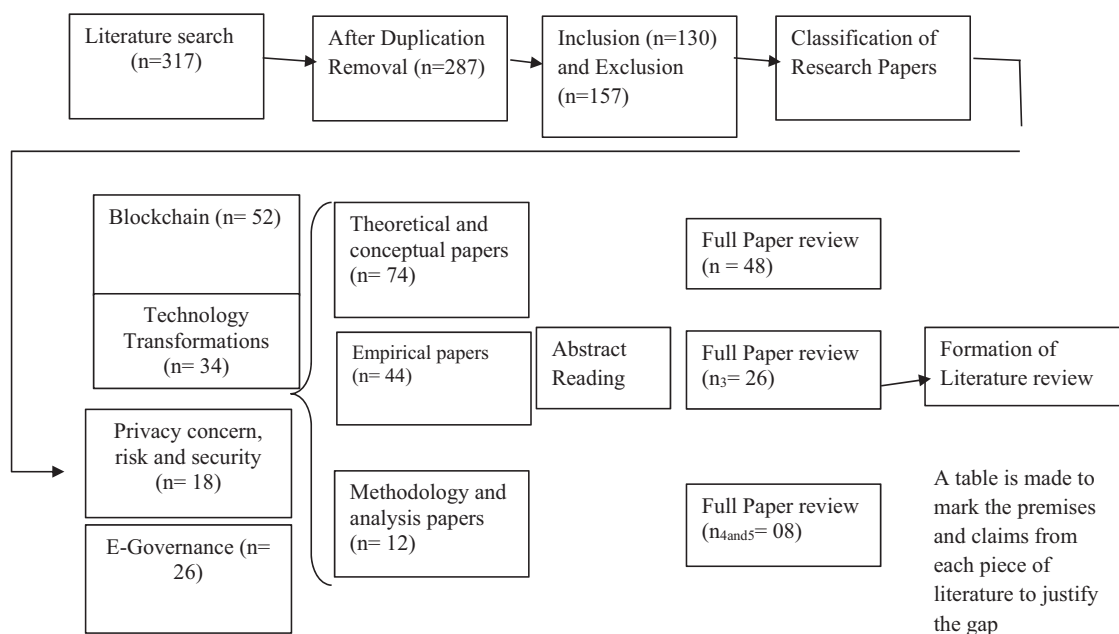


Fig. 1 Overview of the literature review process.

review process (See Fig. 1 below) followed the steps focusing on publications between 2017 to 2023: (a) material identification, (b) sorting the duplication, (c) inclusion and exclusion criteria (d) classification of articles, (e) abstract reading and (f) full content reading.

A list of keywords was prepared following the review of papers through backwards integration (noting the terms that appear multiple times in a set of review papers, identifying their source and following the same process backwards).

Roadmap and conceptualization - bibliometric process. A total of 1123 articles were received initially from databases (Scopus, WoS, and ProQuest) and after applying the limits such as document type – journal publication & review papers, language – English, publication stage – final, subject area – business management, engineering and technology, public governance, economic econometrics and finance which ultimately resulted in 317 articles. Among these 317 duplicate articles (same articles appearing in two or more databases during search with the keywords) and the articles not matching with requirements of the study (might not discuss in detail the link between blockchain and e-governance) were removed, and finally 287. After excluding the irrelevant documents, around 130 articles were taken for bibliometric analysis using the Vosviewer software tool. Though the motive of this study is not to perform bibliometric analysis but to obtain certain details such as top document published, citations and keywords from the analysis would help the research in strengthening the gap, the scientific landscapes and networks of co-citations, co-occurring keywords, a list of top journals, country-wise publications, list of top authors with the latest articles in the field are created to understand the publication trends and future direction Kumar et al. (2023) in Blockchain and E-governance research.

The Fig. 2 above shows the publication trend, the top authors along with year-wise publication analysis (Kumar and Ahmed, 2023). The leading author Salah et al. (2019) with 5 documents and 178 citations in the paper titled “Blockchain for AI: Review and open research challenges” followed by Lo et al. (2019), with 4 documents and 404 citations in a paper titled “Analysis of blockchain solutions for IoT: A systematic literature review”. Other

leading authors are Paik et al. (2019) with 3 documents and 77 citations in the paper titled “Analysis of data management in Blockchain-based Systems: From Architecture to Governance” and Gai et al. (2019) with 3 documents and 468 citations with a paper titled “Privacy-preserving energy trading using consortium blockchain in smart grid”. Recently the trend in publishing articles related to blockchain technology has increased and there are numerous reasons related to it like (1) There is an increase in the requirement of data security in e-governance, (2) advantages provided by blockchain technology related to transparency and safety and (3) convenience for governance companies/institution in providing services to masses accurately. It is also to be noted that there is no significant contribution in existing literature connecting blockchain technology and its benefits for e-governance and digital identity management. Further, studies have not shown the significance of the gig economy and service innovation models in increasing the efficiency and security of e-governance applications.

Publication trends among the countries revealed that with 44 documents, the United States ranked as the country with the highest number of documents published and citations of 1805, followed by China (37 documents, 929 citations), India (31 documents, 584 citations) and the United Kingdom (29 documents, 654 citations) respectively. Year-wise publication trend was covered from 2017 to 2023 as the technology was addressed and discussed, incorporating its benefits for e-commerce lately, after other sectors. Keyword co-occurrence exhibits the list of keywords that were highly occurring across the literature from seminal papers to current publications. The keyword analysis helps in understanding the conceptual connections earlier studies and new connections that can be proposed for the future. This also signifies that the emphasis was not much given by existing literature on incorporating blockchain technology, service innovation models and the gig economy in the DIMF in e-governance.

Blockchain technology. Blockchain is a distributed ledger technology, meaning that the ledger is maintained by a network of nodes instead of a central authority. Once data is recorded on the blockchain, it cannot be altered or deleted, creating a

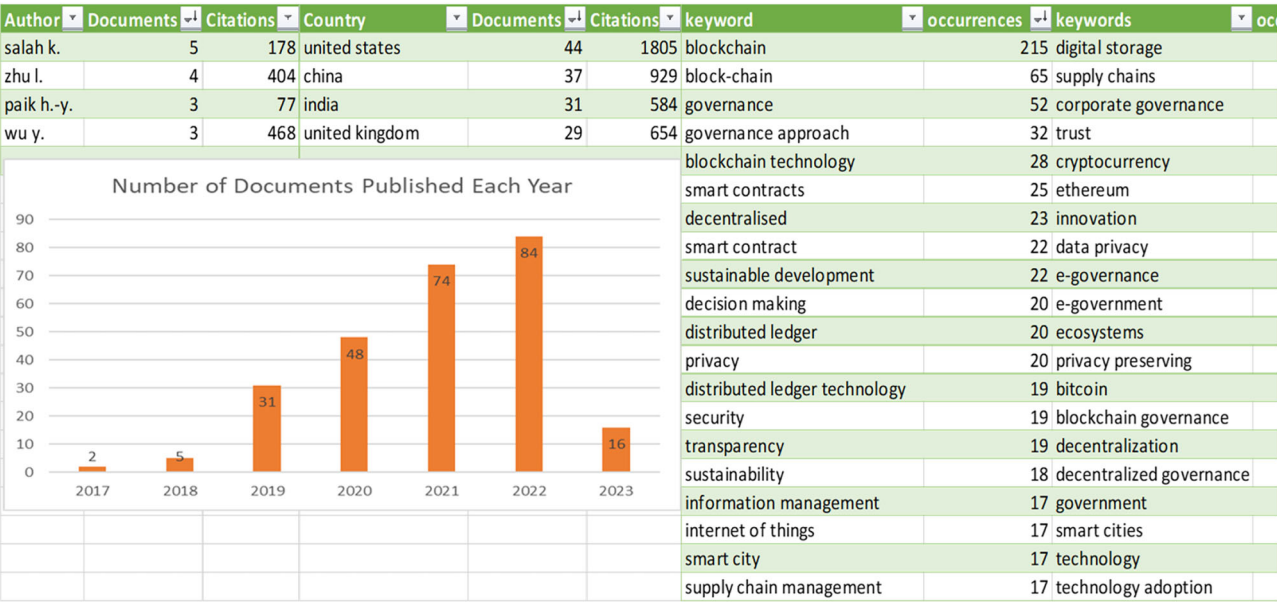


Fig. 2 Blockchain in e-governance publication and top-occurring keywords.

tamper-proof system that provides a high level of security and trust (Yang et al., 2020). All transactions are public, allowing anyone to see the data. This creates a high level of transparency, which can increase trust in the system and prevent fraud. Blockchain uses advanced cryptographic techniques to ensure that transactions are secure and tamper-proof. It relies on a consensus mechanism to ensure that all nodes on the network agree on the current state of the ledger (Viriyasitavat and Hoonsopon, 2019)

Smart contracts are self-executing contracts with the terms of the agreement directly written into code, allowing for the automation of complex processes and new forms of decentralized applications (Koulu, 2016). Blockchain technology can integrate with other technologies and platforms, making it easier to develop new applications and services and integrate them into existing infrastructures (Sharma and Sharma, 2022). Blockchain can automate processes and reduce the need for intermediaries, leading to increased efficiency and cost savings. Some blockchains can handle many transactions per second, making them suitable for enterprise-level applications. Blockchains can tokenize assets, such as stocks, bonds and real estate, allowing them to be traded in a decentralized and secure manner (Benedetti and Rodríguez, 2023).

The key technologies behind the blockchain are (a) Public key cryptography: which encrypts digital identity information using a public key and decrypts it using a private key. Ensures that only authorized parties can access data, (b) Hash functions: Verifies the integrity of digital identity information by hashing each block of the blockchain and including the hash of the previous block, (c) Proof of work (PoW): Used as a consensus mechanism to verify transactions and add them to the blockchain. Can be computationally intensive for e-governance applications (d) Proof of stake (PoS): An alternative consensus mechanism to PoW that requires network nodes to hold a certain amount of cryptocurrency to verify transactions (e) Machine Learning (ML): Used for fraud detection and to improve the accuracy of identity verification processes (Choi and Lee, 2017). Analyzes patterns and detects anomalies in large datasets (f) Smart Contracts: Self-executing contracts that automatically execute transactions based on predefined conditions (Nzuva, 2019). Automates identity verification processes and (g) Decentralized Applications (DApps): Applications that run on decentralized networks, such as blockchain. Used for identity verification and management.

The above Fig. 3 depicts the technology behind blockchain and how blockchain technology is used by various bands in both the private and public sectors. There are three types of blockchain (a) Public Blockchain, (b) Consortium Blockchain and (c) Private Blockchain (Yeasmin and Baig, 2019). A public blockchain is a decentralized network that is open to everyone. Any user can read, write and participate in the blockchain network. It is secured through complex cryptographic algorithms that ensure the integrity of the data stored on the blockchain. A consortium blockchain is a permissioned network that is operated by a group of organizations that work together to maintain and secure the blockchain. It is used for specific business purposes and is open only to the members of the consortium. A private blockchain is a permissioned network that is operated by a single organization or entity. It is used for internal business purposes and is not open to the public. The network participants are controlled and authorized by the organization or entity.

The major limitations of blockchain, as elaborated by (Yli-Huumo et al., 2016; Dai et al., 2017; Wadhwa, 2022), are that firstly, blockchains are limited in their scalability, and this is particularly important for digital identity management. As more people use digital identities, the number of transactions that need to be processed will increase, which can lead to slower processing times and higher fees. Secondly, blockchains offer a high level of security; they can also be seen as a public ledger, which can create privacy concerns. This is particularly important for digital identity management, where the information stored on the blockchain needs to be kept confidential (Naik and Jenkins, 2020). Thirdly, there are many different blockchains available, and they do not always work well together. This can create challenges for digital identity management, where interoperability is critical to ensure that identities can be used across different platforms and services (Bertino et al., 2009; Trivedi, 2023). Fourthly, blockchain technology can be complex and difficult to understand for non-technical users. This can create challenges for digital identity management, where users need to be able to easily manage their identities without needing a deep understanding of the underlying technology (Manski, 2020). Finally, there is still a lack of clear legal frameworks for blockchain technology in many countries. This can create challenges for digital identity management, where legal issues such as liability and dispute resolution need to be addressed (Smedinghoff, 2012).

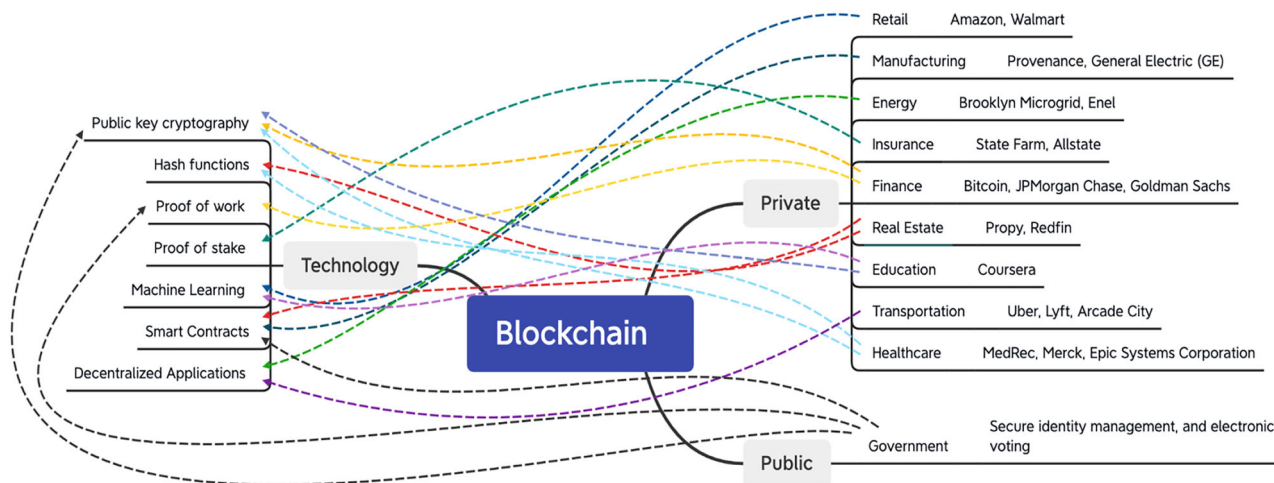


Fig. 3 Framework of blockchain and its role in the Private and Public Sectors.

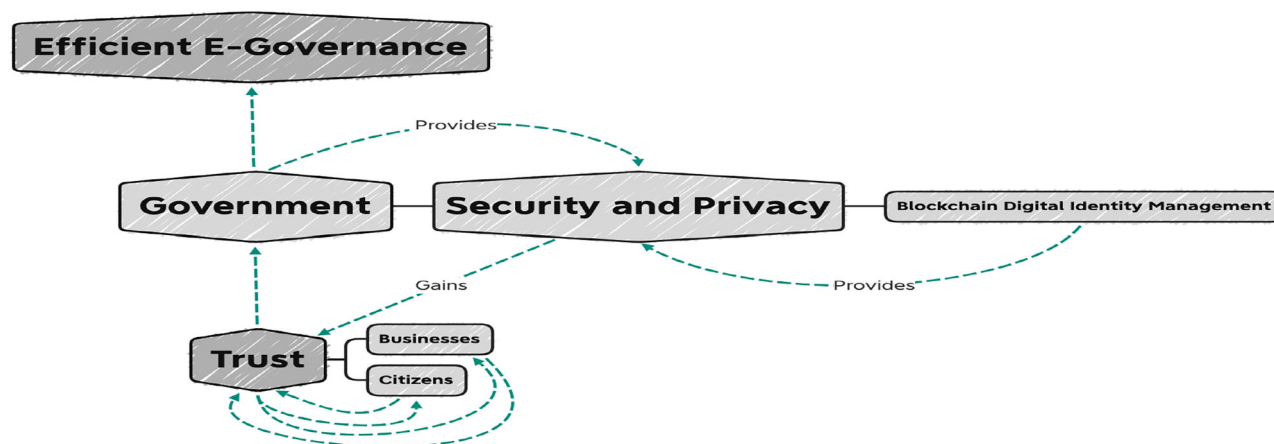


Fig. 4 Existing DIMF integrated with Blockchain Technology.

Digital identity management framework and e-governance.

The DIMF is the collection of information, policies, systems and frameworks of an individual or an organization, helping in the identification process and differentiating from others online. It helps in the context of verification, approvals, transactions and authentications, and hence the tool has to be highly secure and unique, and the probability of error must be minimal. The key components of the DIMF are (a) identity creation like individual detail capturing, (b) Identity verification like biometric authentication/government issued, (c) Identity authentication like a password or smart cards, (d) Identity management through centralized system and (e) Identity usage across multiple systems (Naik and Jenkins, 2020). Digital identity systems are used and highly appreciated by organizations and businesses to track employee details, B2B business information, customer relationship management, patient data handling, student record maintenance etc (Wang, 2023). However, it is easy to handle, rectify, manage and process the DIMF for business organizations and service sectors as the quantity of data is lesser and the quality of information is high and complete compared to the government organizations/agencies (Fig. 4). There need for a highly protective, secure, efficient and enhanced E-Governance digital identity framework with the latest tools and technologies (Dawes, 2008). Figure 5 below depicts the existing identity management IOT using blockchain and smart contracts.

The implementation of the latest technology and management tools, along with blockchain technology (Fig. 6) is a need

of the hour to enhance DIMF for smooth communication between the administration and the citizens. This offers and allows every individual in the respective county to access the public services freely with transparency (Dawes, 2008). Employing DIMF to supervise resources and administer policies, results in a citizen-directed system (Palvia and Sharma, 2007). E-governance provides unshrinking access to government services and information around the clock every day to users, and hence governments need to restructure their operations to improve their service delivery to the point where it is efficient, safe, and secure. In addition, public participation in decision-making helps the government to achieve the implementation of its services to the next level (Bannister and Connoiiy, 2019).

Data protection becomes vital and challenging when public services are provided online. Hence, it becomes necessary to develop both systems and models that provide and improve the security of e-government. Speed, services at a low cost, transparency and accountability are the advantages, whereas huge infrastructure investments, a loss of interpersonal communication, illiteracy and cybercrime are the challenges faced by e-governance.

Methodology

The issues were articulated by integrating the frameworks depicted in Fig. 4. This integration yields a conceptual model, which is then subjected to further validation through expert interviews and fuzzy logic. The validation process is

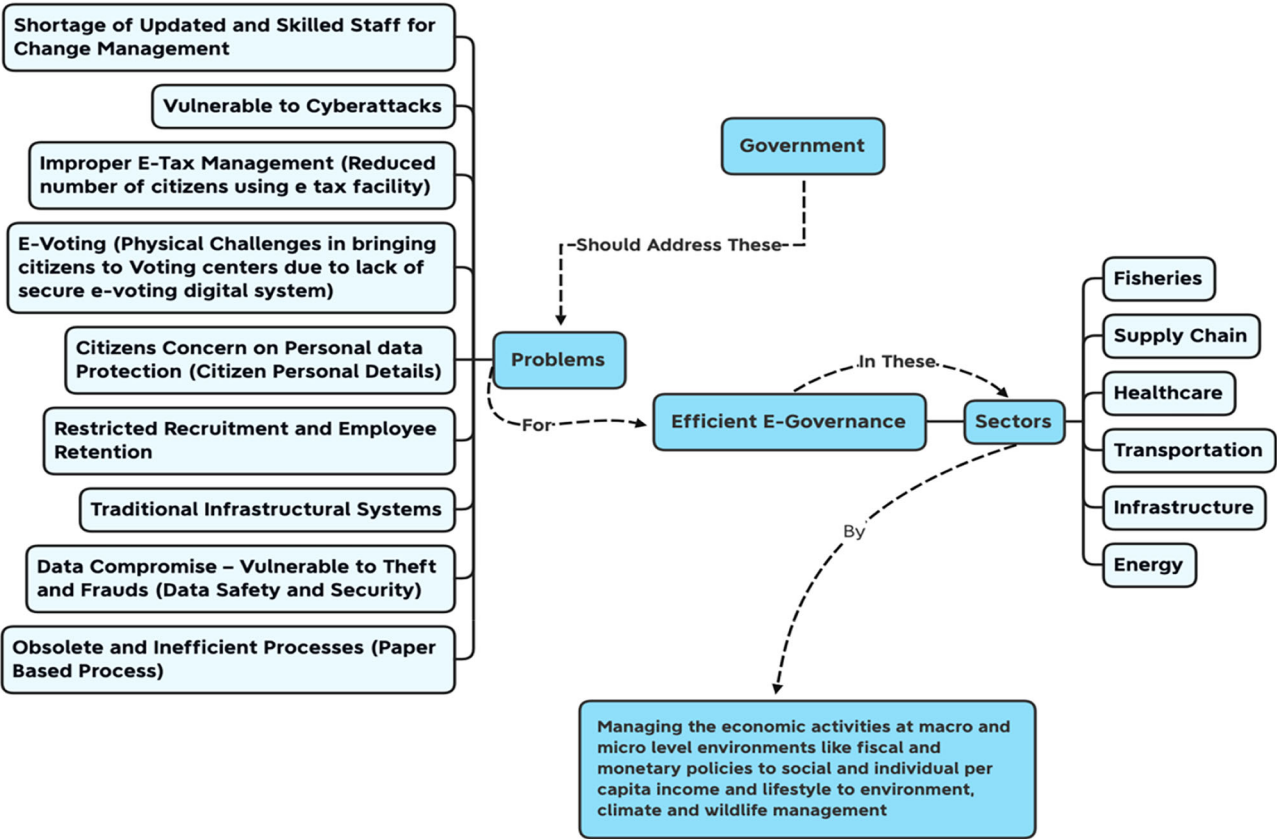


Fig. 5 Existing issues in Digital identity management framework in e-commerce.

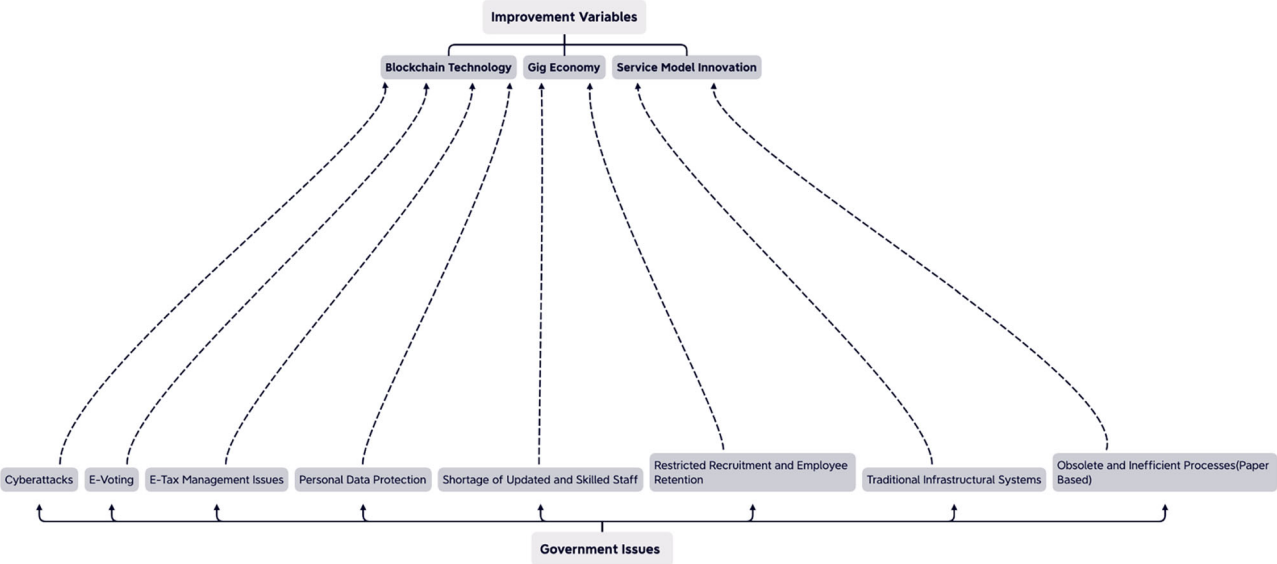


Fig. 6 List of Improvement variables to address the existing issues in DIMF E-Governance.

comprehensive and involves applying the opinions of experts, as well as validating the rules used for the fuzzy inference system through a structured questionnaire that employs a semantic differential scale to obtain a more profound understanding of opinions. The primary advantage of this approach is that it simplifies the scenario design process, making it more applicable and increasing the likelihood of success. The fuzzy logic theory has broad applicability in various fundamental domains, such as sociology, medicine, finance, and management, specifically in

decision-making. The fuzzy logic-based decision system proposed in this study is capable of validating the proposed model (Ji et al., 2007). To begin with fuzzy logic, the first step is to define the input variables and their corresponding membership functions. Gaussian functions are used to define the membership functions for the input variables ‘Blockchain,’ ‘gig economy,’ and ‘Service model innovation.’ The mean and standard deviation parameters of the Gaussian functions are established based on the data and

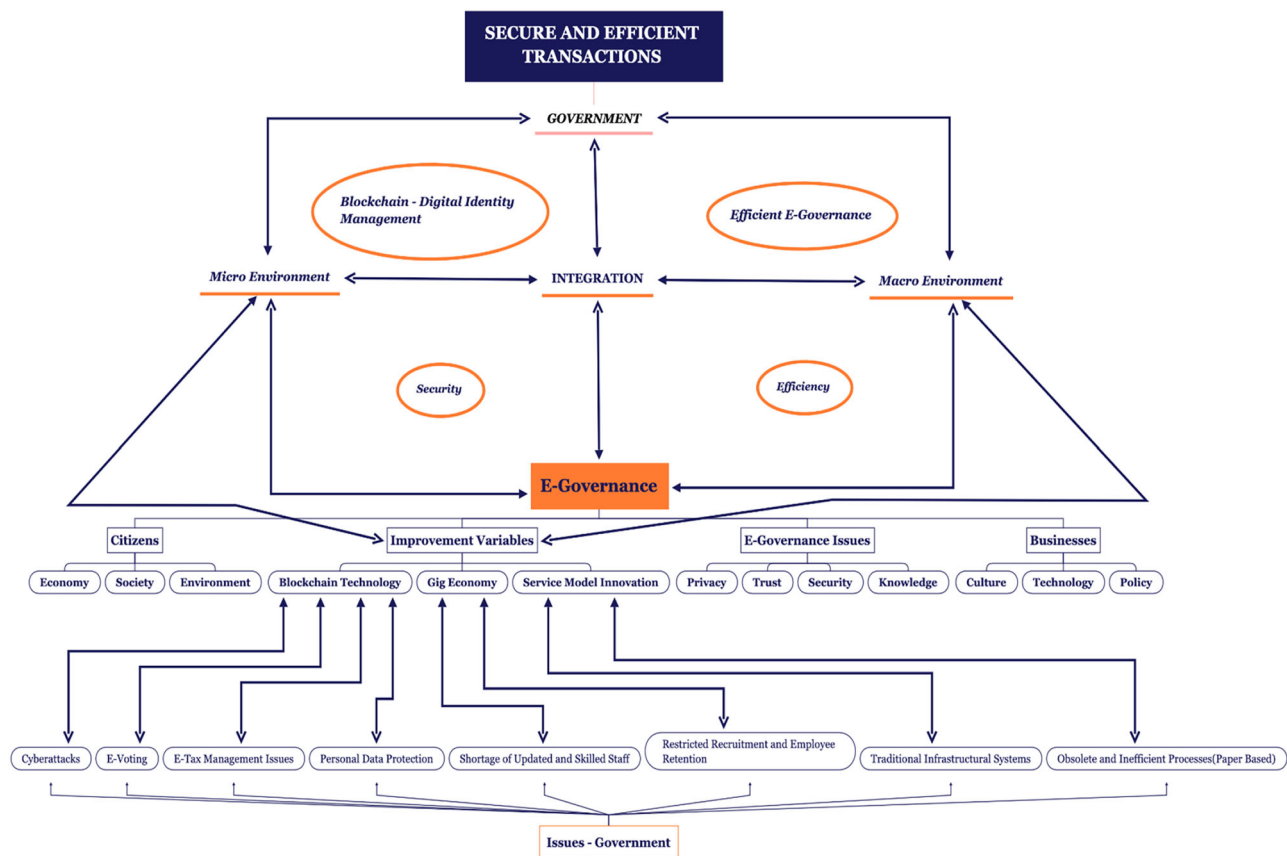


Fig. 7 Proposed enhanced blockchain-digital identity management model.

the problem domain acquired from the questionnaire. Next, the output variable 'Secure and Efficient Transactions' is defined, along with its corresponding membership function, which is triangular in this case (Yadav et al., 2021). The parameters of the triangular function, such as the left and right slopes and the peak, are set according to the desired output. After defining the input and output membership functions, the next step involves establishing the fuzzy rules that represent the mapping between the input and output variables. The fuzzy rules can be defined using linguistic terms like "if Blockchain is high and the gig economy is high and Service model innovation is high, then Secure and Efficient Transactions is very high" (Babkin, 2018). The number and complexity of the fuzzy rules will depend on the problem domain. Once the input and output membership functions, along with the fuzzy rules, are established, the fuzzy inference system can be employed to evaluate the proposed model (Wang et al., 2022). The system maps crisp input values to fuzzy values using the membership functions, applies fuzzy rules to determine the degree of membership of the output variable and then defuzzifies the output variable to obtain a crisp value (Bai and Wang, 2006). To evaluate the performance of the model, the predicted values are compared with the actual values using metrics such as mean squared error and with the author's predicted values. As the performance was deemed satisfactory, no modification to the input and output membership functions or the fuzzy rules was necessary, as the desired level of performance was achieved.

Enhancing secure and efficient transactions: integrating blockchain, service innovation business model & gig-economy. The proposed blockchain-based e-government framework is illustrated in Fig. 7 below, which integrated the concepts of the

gig economy and service innovation models to enhance the effective, efficient and secure transactions in the e-governance context. The figure indicates the importance of an interaction that needs to take place between government departments, organizations and individual citizens, allowing a peer-to-peer (p2p) exchange (broadcast) and validation of data being submitted by individuals. The information sharing between citizens and the government, such as filling out forms for tax, marriage certificates, business permits, birth certificates, visas or passports, is the process of identity creation, verification, authentication and validation (Elisa et al., 2018).

The gig economy is the conceptual framework in the management domain that explains the trend of hiring and managing contractors and freelancers on an adjunct/part-time/hourly basis for short-term projects, managing the irregular capacity in services. It can play several beneficial roles in e-governance, especially in digital identity management. Identification of specialized talent that is cost-efficient with the increasing use of technology and digitalization. The gig economy can help the government tap into a pool of highly skilled independent contractors with expertise in certain areas (Graham et al., 2017). Can assist government agencies in providing services more efficiently, especially in public transport or parking services. Citizens can also participate in government jobs on a timely basis to improve the service infrastructure. On integrating with blockchain technology, the gig-economy framework would enhance cost-efficient service outcomes (Patwa et al., 2021).

Service innovation is the process of creating new or improved services to meet the changing needs and requirements of the market and customers. In emerging economies, there are multiple gaps, problems, and opportunities to be addressed on a day-to-day basis to improve the comfort, lifestyle and requirements of

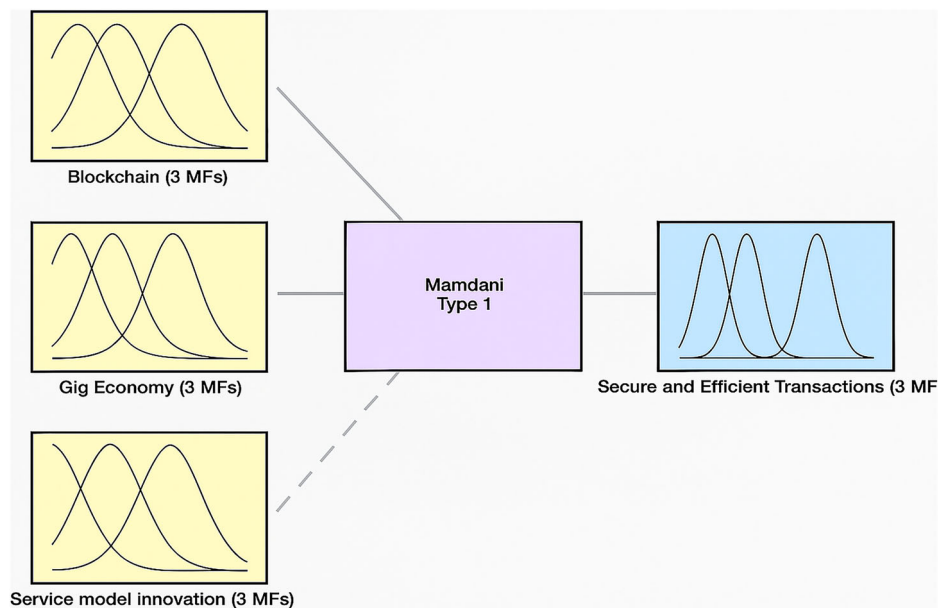


Fig. 8 Fuzzy Inference System.

the people (Basu, 2004). To be precise, for government services, there are always insufficient resources, high costs of services, growing demand and security issues. In e-governance, service innovation can be used to improve service delivery, service communication and policy appropriate to the requirements of the citizens (Bokhari and Myeong, 2022). There needs to be frequent and consistent innovation to improve convenient, accessible and user-friendly frameworks along with secured transactions through blockchain. Hence, this study developed a framework that could enhance digital identity management in existing blockchain-enabled e-governance applications.

Estimation of the path for the proposed model

Simulation result. The mean and standard mean of each variable, blockchain (6.6667,1.68034), gig economy (6.2222,1.35280) and service model innovation (6.0000,1.68034) were applied for defining the membership functions of the inputs and for the output of secure and efficient transactions (7.1111,75840). The Mamdani type 2 inference system has simulated 27 possible rules (see Appendix 1). The fuzzy logic toolbox in MATLAB has features to check for efficiency by changing the values of output and input variables to obtain the desired result. After quantifying the logic of the rule created, the proposed model proved to be efficient. The validation of the research considered the formulation of a scale and formula generation for the input variable to enhance the performance of the model. The fuzzy inference system, rule inference and control surface map used for the model are shown in Figs. 8–10 below.

Scale and formula calculation and results

To calculate the impact of service model innovation for secure and efficient transactions in e-governance the following steps were adopted. Step 1: Identify the key metrics that define service model innovation, secure transactions and efficient transactions in e-governance. These could include factors such as:

Service Innovation: Number of new services introduced, user satisfaction with the new services, reduction in processing time for services, increase in accessibility and convenience of services.

Secure Transactions: Number of successful transactions, number of fraudulent transactions detected and prevented, level of user confidence in the security of the system, level of compliance with data privacy and security regulations.

Efficient Transactions: Average processing time per transaction, percentage of transactions completed without errors, reduction in administrative costs, reduction in time required to resolve disputes.

Step 2: Assign weights to each of these metrics based on their relative importance. For example, the number of successful transactions may be more important than user satisfaction with new services, so it could be assigned a higher weight.

Step 3: Calculate the impact score for each metric by multiplying its weight by the change in value observed after implementing the service model innovation. For example, if the average processing time per transaction was reduced by 30% and this metric had a weight of 0.3, the impact score for this metric would be $0.3 \times 0.3 = 0.09$.

Step 4: Sum up the impact scores for all metrics to obtain a total impact score for the service model innovation. This score can be used to compare the effectiveness of different innovations or to track the impact of the same innovation over time.

To derive a scale for measurement, assign numerical values to the impact scores based on their magnitude. For example, the following scale:

Low Impact: 0–0.3
Moderate Impact: 0.4–0.7
High Impact: 0.8–1.0

Using this scale, categorization can be done to determine the impact of different service model innovations and compare their effectiveness.

To calculate the impact of the gig economy on secure and efficient transactions in e-governance. Step 1: Identify the key metrics that define the impact of the gig economy on e-governance.

These could include factors such as:

- Number of gig workers engaged
- Reduction in administrative costs

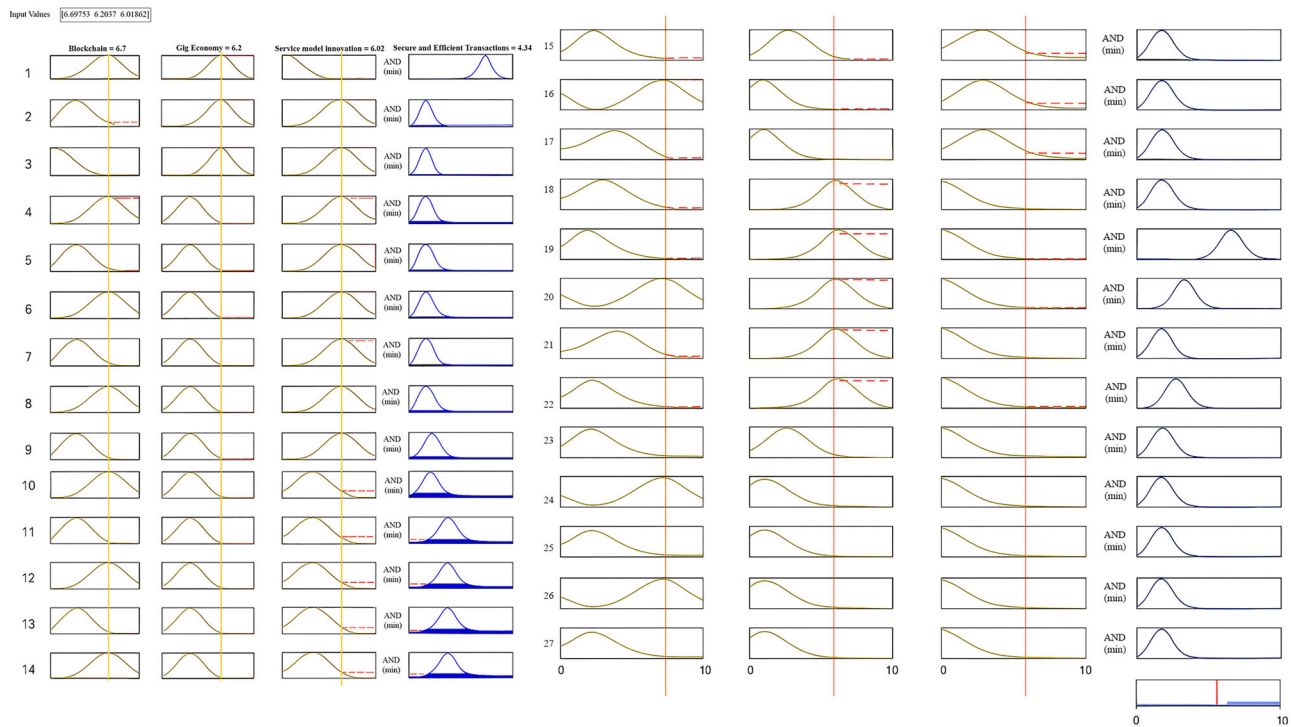


Fig. 9 Rule Inference.

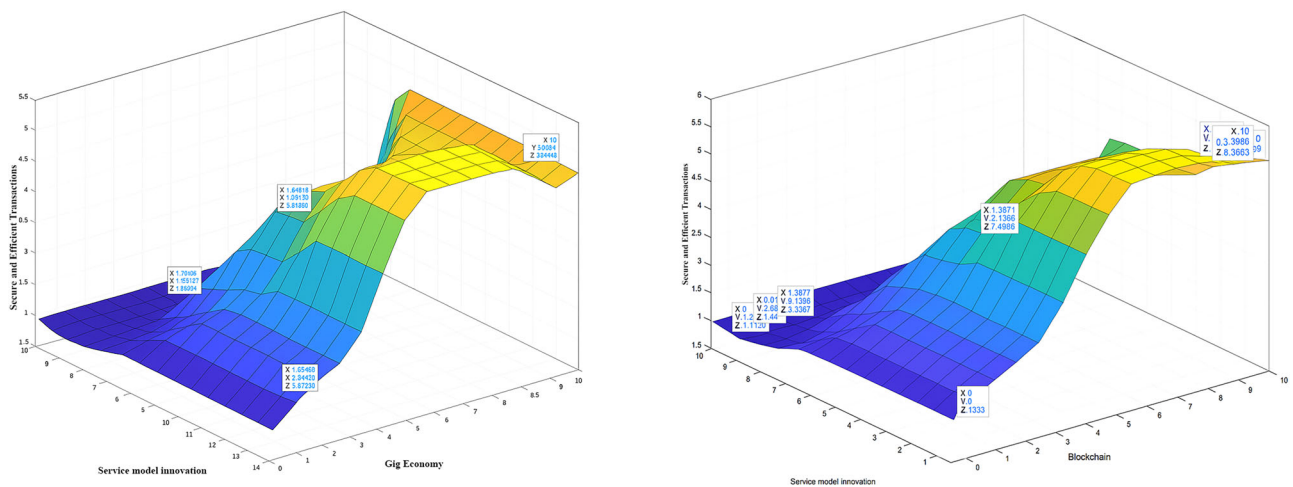


Fig. 10 The control surfaces 1 & 2 outline the graphical representation of the results. Detailed MATLAB code with analysis is given in Appendix 1.

- Increase in accessibility and convenience for citizens
- User satisfaction with the services provided by gig workers
- Level of compliance with data privacy and security regulations

Step 2: Assign weights to each of these metrics based on their relative importance. For example, the number of gig workers engaged may be more important than user satisfaction with the services provided, so it could be assigned a higher weight.

Step 3: Calculate the impact score for each metric by multiplying its weight by the change in value observed after implementing the gig economy model. For example, if the reduction in administrative costs was 20% and this metric had a weight of 0.4, the impact score for this metric would be $0.4 \times 0.2 = 0.08$.

Step 4: Sum up the impact scores for all metrics to obtain a total impact score for the gig economy model. This score can be used to compare the effectiveness of different gig economy models or to track the impact of the same model over time.

To derive a scale for measurement, assign numerical values to the impact scores based on their magnitude. For example, the following scale:

- Low Impact: 0–0.3
- Moderate Impact: 0.4–0.7
- High Impact: 0.8–1.0

Using this scale, categorization can be done to determine the impact of different gig economy models and compare their effectiveness.

To calculate the impact of blockchain for secure and efficient transactions in e-governance. Step 1: Identify the key metrics that define the impact of blockchain on e-governance. These could include factors such as:

- Number of transactions processed
- Reduction in transaction processing time
- Reduction in administrative costs
- Increase in transparency and accountability
- Level of compliance with data privacy and security regulations

Step 2: Assign weights to each of these metrics based on their relative importance. For example, the reduction in transaction processing time may be more important than the level of compliance with data privacy and security regulations, so it could be assigned a higher weight.

Step 3: Calculate the impact score for each metric by multiplying its weight by the change in value observed after implementing the blockchain system. For example, if the reduction in administrative costs was 30% and this metric had a weight of 0.3, the impact score for this metric would be $0.3 \times 0.3 = 0.09$.

Step 4: Sum up the impact scores for all metrics to obtain a total impact score for the blockchain system. This score can be used to compare the effectiveness of different blockchain systems or to track the impact of the same system over time.

To derive a scale for measurement, assign numerical values to the impact scores based on their magnitude. For example, the following scale:

- Low Impact: 0–0.3
- Moderate Impact: 0.4–0.7
- High Impact: 0.8–1.0

Using this scale, categorization can be done to determine the impact of different blockchain systems and compare their effectiveness.

Note: It's important to note that this scale is just an example, and the actual values and categories may vary depending on the specific context and requirements of the e-governance system.

The formula was generated from experts' opinions and used business mathematics about the proposed model in the study.

The fuzzy inference system shows the inputs blockchain, gig economy, and service model innovation as inputs and secure and efficient transactions as output. The Mamdani Type 1 is the fuzzy inference system where the rules are fired for optimal results. The outcomes signify a structured approach to improving secure and efficient transactions within e-governance systems. Beginning with the integration of frameworks and the development of a conceptual model, the process emphasizes simplifying scenario design and increasing the success rate in e-governance transactions. Integrating the framework depicted in Fig. 4 and subjecting the resulting conceptual model to validation through expert interviews and fuzzy logic, the study ensures a foundation for subsequent analysis and implementation. Through the application of Gaussian and triangular functions for input and output variables, respectively and the establishment of fuzzy rules based on linguistic terms, the system effectively evaluates the proposed model. The simulation and validation process, facilitated by MATLAB's fuzzy logic toolbox, further solidifies the model's efficiency by analyzing various input and output variables and confirming satisfactory performance metrics. The estimation and calculation of impact provide insight into the effectiveness of key components of the service model innovation, the gig economy, and blockchain technology. Identifying and weighting key metrics associated with each component, impact scores are calculated and categorized to assess their respective contributions to secure and

efficient transactions within e-governance. This methodical approach enables stakeholders to gauge the effectiveness of different strategies and prioritize interventions accordingly. The interpretation of results, particularly demonstrated through the fuzzy inference system in Fig. 8, underscores the interconnectivity of inputs, blockchain, the gig economy, and service model innovation with secure and efficient transactions. The Mamdani Type 1 fuzzy inference system optimizes output by firing rules based on input variables, highlighting the dynamic nature of the model and its responsiveness to varying conditions within e-governance environments (see Appendix 1). Subsequently, the outcomes signify a structured and rigorous methodology for enhancing secure and efficient transactions in e-governance, integrating frameworks, fuzzy logic, and expert validation to optimize performance and drive meaningful outcomes in e-governance settings.

The rule inference in the figure elaborates on the values assigned. In MATLAB, the vertical lines can be scrolled left or right until the desired value of output is obtained from the model.

Discussions

As for the government services, there were insufficient resources, high costs of services, growing demand, and security issues. This prompted a study that delves into the integration of blockchain technology, gig economy practices, and service innovation models within e-governance frameworks, aiming to enhance efficiency, security and accessibility. The key findings highlight the potential of this integration to revolutionize digital identity management and transaction processing in various e-governance sectors (like healthcare, safety and security, public welfare schemes etc.). The result helped in calculating the scaled measurements and supports the government's deal, ensuring secure and efficient transactions, ensuring citizen satisfaction. As a consequence of digital cyber-attacks, the occurrence of fraud cases in financial transactions, preparation of fake documents for government benefits, it is difficult for citizens to trust or for the government to establish confidence among their service receivers. Blockchain would be the best option to resolve this issue, however, the limitation of the blockchain explained earlier indicates that the technology cannot manage the transaction at a larger scale (Yli-Huumo et al., 2016 and Dai et al., 2017). Leveraging blockchain's decentralized ledger system, governments can establish secure and transparent DIMFs, mitigating privacy concerns and ensuring data integrity. Furthermore, the study indirectly addresses the risks associated with different blockchain functions and their potential impact on data integrity and property loss. The interoperability issues between disparate blockchain networks, as noted in the study, can exacerbate these risks, particularly in scenarios involving the transfer of sensitive data or assets across chains. In light of these challenges, the study's exploration of blockchain integration within the broader context of digital identity management underscores the importance of addressing scalability, privacy, and interoperability concerns. Acknowledging these issues and proposing practical solutions, such as fuzzy logic-based decision systems and integrated frameworks, the study contributes to the ongoing discourse on optimizing blockchain technology for secure and efficient digital identity management within e-governance ecosystems.

In digital identity management, the study highlights blockchain technology's provisions on a decentralized and secure framework, ensuring the integrity and privacy of citizen's digital identities. As digital identity transactions escalate, the demand for processing, validation, and storage surges correspondingly. This can strain existing infrastructure, resulting in slower processing times and increased operational costs, which aligns with the study's

findings. The gig economy's ability to tap into specialized talent pools and adapt to fluctuating service demands offers governments a scalable solution to address resource constraints and meet evolving citizen needs. Subsequently, governments can establish tamper-proof identity systems, bolstered by the gig economy's flexible workforce for streamlined identity verification processes. Service innovation models in the study's framework further enhance accessibility and usability, ensuring citizens can easily interact with digital identity platforms. Moreover, in transaction processing, blockchain's distributed ledger system enables transparent and efficient transactions, such as payment processing and license issuance. Furthermore, the study's exploration of gig economy principles underscores the potential for flexible workforce arrangements to optimize resource allocation and support transaction tasks, improving the speed and efficiency of service delivery within government agencies, and mirroring successful implementations in companies like Uber and Airbnb.

Subsequently, by tapping into a pool of skilled independent contractors and freelancers, government agencies can address staffing gaps, improve service accessibility, and foster innovation in service provision. Additionally, blockchain ensures data integrity and transparency, providing governments with secure and immutable records of transactions and public information. Moreover, service innovation emerges as a critical enabler of e-governance effectiveness, enabling governments to tailor service offerings, streamline communication, and enhance policy responsiveness. The study highlights the imperative for continuous innovation to address resource scarcity, high service costs, and burgeoning demand, particularly within government sectors where efficiency and accessibility are paramount. Embracing emerging technologies and management paradigms, governments can transcend traditional bureaucratic barriers, fostering greater citizen engagement and satisfaction. Service innovation models can further enhance data accessibility and transparency by introducing user-friendly interfaces and communication channels. Adopting agile and user-centric approaches to service design and delivery, governments can adapt to evolving citizen needs and preferences, driving continuous improvement and innovation in public service provision. The fuzzy inference system, employed in the study, demonstrates the efficacy of integrating blockchain, gig economy, and service innovation inputs to optimize secure and efficient transactions, offering governments a roadmap for effective implementation. Quantifying the statements and opinions yielded more scalable and applicable results.

E-government initiatives increasingly rely on advanced digital infrastructures to deliver public services efficiently, securely, and transparently. At the core of this transformation lies blockchain technology, offering immutable records and decentralized control that can enhance data privacy and trust in digital governance. With rising concerns around identity theft and data breaches, blockchain's cryptographic protocols ensure enhanced privacy protection for citizens' digital identities. Moreover, the emergence of digital currencies and cryptocurrencies, including Central Bank Digital Currencies (CBDCs) (Alonso et al., 2025), further bolsters e-governance frameworks by enabling secure, transparent, and traceable financial transactions. CBDCs, provide governments with the ability to maintain monetary sovereignty while facilitating real-time payments and reducing transaction costs (Zarifis and Cheng, 2025; Chua et al., 2025). Collectively, these technologies offer a future-ready foundation for redefining trust, transparency, and operational efficiency in public service ecosystems.

The findings reveal the significance of this research within the broader literature on e-governance, digital identity management and emerging technologies. Previous studies have explored the

individual components of blockchain, the gig economy and service innovation, but this research contributes by examining their synergistic effects within e-governance contexts. Service innovation models introduce digital platforms and mobile applications, streamlining service delivery processes and enhancing accessibility for citizens. Overall, the integration optimizes e-governance processes, leading to increased efficiency, security, and accessibility for citizens and government agencies alike. Unexpected results may arise in the form of challenges during the integration process. For example, issues related to data privacy, scalability of blockchain networks, and adaptation of gig economy practices to government workflows may surface. These results underscore the complexity of implementing integrated solutions and highlight the need for comprehensive planning and stakeholder engagement. Addressing limitations in the research is essential for maintaining credibility. Challenges include the dynamic nature of technology adoption. Acknowledging these encourages future researchers to build upon this study's findings and address gaps in knowledge. Potential follow-up research studies could explore the long-term impacts of integrating blockchain, gig economy practices and service innovation models in diverse e-governance settings. Additionally, investigations into regulatory frameworks, user acceptance and cybersecurity measures could provide valuable insights into optimizing the integration process.

Implications

The government has notable implications from the outcome of this research, where they can understand the level of integration needed for the highest effective transaction of their service to larger audiences. The study also identified the specific areas in each construct, such as service innovation, the gig economy and blockchain, that needed to be considered to manage digital identity safely and securely and at the same time provide enhanced services. Socially, the manuscript contributes to the government and citizens of developing and underdeveloped countries to efficiently use blockchain technology, available resources, and reverse innovation techniques to optimize and maximize the outcome. Further, it can be understood and confirmed that blockchain is the key technology that could ensure the safety and security of the data with high transparency and convenience. Thus, the government must pursue the technology availability and implement it in their streams to better serve their citizens. The integration of the gig economy and service innovation models presents promising avenues for optimizing government services and resource management. Recruiting specialized talent and leveraging innovative service delivery methods, governments can enhance service accessibility, efficiency and user satisfaction.

Academic implications. The academic implications of the research are multifaceted and contribute to the advancement of knowledge in several key areas. Firstly, the study provides a synthesis of theoretical frameworks related to digital identity management and e-governance. Examining existing literature and integrating concepts from diverse disciplines such as technology, management and public administration, the research offers a comprehensive understanding of the complex dynamics at play in modern governance systems. Scholars have the opportunity to explore these insights for crafting robust methodologies aimed at scrutinizing digital identity management and e-governance across a spectrum of settings. These settings encompass diverse landscapes, including but not limited to varying countries, regions, and cultural milieus. Additionally, scholars can explore the socioeconomic dimensions, urban and rural environments, to ascertain the nuanced impact of digital identity management and

e-governance. Furthermore, investigations may delve into the structural dynamics of different organizations, spanning governmental bodies, private enterprises, and non-profit entities. Also, a probe of the evolving technological landscapes, considering stages of adoption and the interplay with regulatory frameworks and legal structures. Scrutinizing these varied contexts, scholars aim to elucidate the implementation of digital identity management and e-governance practices, their alignment with local norms and regulations and their ramifications on individuals, organizations, and broader societal constructs. Theoretical frameworks derived from this research help guide future studies, providing a solid foundation for analyzing the interplay between technology, management concepts, and public policy. Moreover, the study inspires further research into innovative approaches to addressing challenges in digital identity security, transaction efficiency, and governance practices. Exploring interdisciplinary collaboration and theoretical synthesis, the research contributes to the enrichment of academic discourse in fields such as information systems, public administration, and management studies.

Practical implications. The practical implications of the research are significant for practitioners in government agencies and public service organizations. Understanding the potential of blockchain technology, the gig economy, and service innovation, government entities can enhance their operational efficiency and service delivery mechanisms. Practitioners can leverage the proposed framework as a practical roadmap for optimizing resource allocation, improving transaction efficiency, and enhancing transparency and accountability in government services. Incorporating blockchain technology, governments can strengthen digital identity security and protect sensitive citizen data from unauthorized access or tampering. Furthermore, embracing the principles of the gig economy and service innovation, government agencies can adapt to evolving citizen needs and preferences, delivering more responsive and user-centric services. The study's findings offer practical insights and actionable recommendations for policymakers and government administrators seeking to modernize their governance systems and enhance citizen satisfaction. Elucidating the benefits and challenges associated with integrating blockchain technology, the gig economy, and service innovation within e-governance frameworks, the research offers practical guidance on optimizing resource allocation, streamlining transaction processes, and fostering transparency. Policymakers can leverage these insights to design robust strategies for digital identity management, ensuring secure and efficient service delivery while bolstering citizen trust and confidence in government initiatives. Through proactive adoption of innovative technologies and management paradigms, policymakers can position their administrations at the forefront of digital transformation, driving positive societal outcomes and enhancing overall governance effectiveness.

Societal implications. The societal implications of the research extend to citizens and communities, shaping the relationship between government institutions and the public. Enhanced digital identity management instils trust and confidence among citizens in government services, fostering a sense of security and reliability in online transactions and interactions with public authorities. Adopting emerging technologies and innovative management concepts, governments can address societal challenges and improve the overall quality of life for citizens. The research highlights the potential for governments to leverage digital identity management systems to promote inclusivity, accessibility, and transparency in service delivery. Moreover, by fostering civic engagement and participation, governments can empower

citizens to actively contribute to decision-making processes and shape public policies that reflect their interests and concerns. The study underscores the transformative potential of technology-enabled governance systems in promoting social cohesion, democratic values, and citizen empowerment.

Conclusion

To conclude, the study indicated the integration of digital identity management, blockchain technology, and e-governance highlights critical challenges and their potential solutions, offering significant benefits across academic, practical, and societal realms. Addressing the inherent scalability and privacy concerns of blockchain technology, the conversation recognizes the growing demand for processing digital identity transactions. However, with the increasing volume of these transactions come challenges related to validation, storage, and privacy preservation. While blockchain's transparent nature ensures robust security measures, it also raises concerns about data privacy and confidentiality. Consequently, there is a pressing need to balance the scalability limitations of blockchain with the imperative of protecting user privacy. The study's integration of blockchain technology into e-governance systems shows promise in enhancing digital identity security and trust in transactions. Nonetheless, challenges arise from the interoperability and integration issues between different blockchain networks. The proposed integration framework offers a structured approach to overcome these hurdles, leveraging the gig economy and service innovation models. This approach optimizes resource allocation, improves transaction efficiency, and enhances service delivery while promoting transparency and accountability.

Despite the potential benefits, the complexity of blockchain technology presents barriers to widespread adoption, particularly for non-technical users. The lack of clear legal frameworks further complicates its implementation in government systems. To address these challenges, user-friendly interfaces and educational initiatives are essential. Governments can empower citizens to meaningfully participate in e-governance processes by fostering digital literacy and simplifying complex concepts. Therefore, the examination of digital identity management, blockchain technology, and e-governance frameworks in the study emphasizes the need for collaborative efforts to overcome existing challenges and capitalize on emerging opportunities. Addressing scalability, privacy, interoperability, and user understanding, stakeholders can harness blockchain technology's transformative potential to improve service delivery, promote citizen engagement, and foster inclusive governance practices. The primary aim of this research is to establish a comprehensive framework for digital identity management, leveraging blockchain technology to enhance security and incorporating gig economy concepts and service innovations for more efficient transactions in e-governance. The study employs fuzzy logic techniques and in-depth interviews with experts to identify the predominant construct and variables. The results highlight the importance of blockchain in capturing, validating, and authenticating the digital identity of citizens. The decentralized and transparent nature of blockchain ensures data integrity, emphasizing the need for awareness among service providers about its benefits for safety and security. The proposed framework benefits governments in both developing and underdeveloped countries, enabling resource management through the gig economy and adapting input variables based on sector requirements through service innovation. Future research can explore predictive algorithms in the e-governance context using this model.

Data availability

No datasets were generated or analysed during the current study.

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Author contributions

Arjun J. Nair contributed to the methodology, writing of the original draft and analysis. Satish Rao A. B. provided resources and secured funding for the study. Sridhar Manohar was responsible for validation and review of the work.

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Competing interests

The authors declare no competing interests.

Ethical approval

The study does not involve human participants or their data, hence, ethics approval was not required.

Informed consent

The study does not involve human participants or their data, hence, informed consent was not required.

Additional information

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